

40.520: Stochastic Models**Assignment 2***Due date: Mar 1st, 2026 before midnight*

Problem 1. Consider two independent Markov chains $(X_n : n \geq 0)$ and $(Y_n : n \geq 0)$ defined on the state space I . Suppose (X_n) has transition matrix P and (Y_n) has transition matrix Q , and their evolutions are independent. Is the process $Z_n = (X_n, Y_n)$ a Markov chain? If so, find its transition probabilities.

Solution. Let $\mathcal{F}_n = \sigma(X_0, \dots, X_n, Y_0, \dots, Y_n)$ be the history up to time n (i.e., the σ -algebra up to time n). Because $(X_n : n \geq 0)$ and $(Y_n : n \geq 0)$ are independent processes, the events $\{X_{n+1} = k\}$ and $\{Y_{n+1} = l\}$ are conditionally independent given $\mathcal{F}_n$. Therefore:

$$\mathbb{P}(X_{n+1} = k, Y_{n+1} = l \mid \mathcal{F}_n) = \mathbb{P}(X_{n+1} = k \mid \mathcal{F}_n) \cdot \mathbb{P}(Y_{n+1} = l \mid \mathcal{F}_n).$$

By the Markov property, $\mathbb{P}(X_{n+1} = k \mid \mathcal{F}_n) = \mathbb{P}(X_{n+1} = k \mid X_n) = P_{X_n, k}$ and $\mathbb{P}(Y_{n+1} = l \mid \mathcal{F}_n) = \mathbb{P}(Y_{n+1} = l \mid Y_n) = Q_{Y_n, l}$. Thus:

$$\begin{aligned} & \mathbb{P}(Z_{n+1} = (k, l) \mid \mathcal{F}_n) \\ &= \mathbb{P}(X_{n+1} = k, Y_{n+1} = l \mid \mathcal{F}_n) \\ &= \mathbb{P}(X_{n+1} = k \mid \mathcal{F}_n) \cdot \mathbb{P}(Y_{n+1} = l \mid \mathcal{F}_n) \\ &= P_{X_n, k} \cdot Q_{Y_n, l}. \end{aligned}$$

Therefore, $\mathbb{P}(Z_{n+1} = (k, l) \mid \mathcal{F}_n)$ depends only on $Z_n = (X_n, Y_n)$, not on earlier history. Hence, the Markov property holds:

$$\mathbb{P}(Z_{n+1} = (k, l) \mid \mathcal{F}_n) = \mathbb{P}(Z_{n+1} = (k, l) \mid Z_n).$$

In addition,

$$\mathbb{P}(Z_{n+1} = (k, l) \mid Z_n = (i, j)) = P_{i, k} \cdot Q_{j, l}.$$

Problem 2. Consider a Markov chain with state space $S = \{A, B, C, D, E\}$ and the following transition rules:

- From A : Go to B with probability $1/2$, go to C with probability $1/2$.
- From B : Go to A with probability $1/3$, go to C with probability $1/3$, go to D with probability $1/3$.
- From C : Go to A with probability $1/4$, go to B with probability $1/4$, go to D with probability $1/4$, go to E with probability $1/4$.
- From D : Go to B with probability $1/2$, go to E with probability $1/2$.
- From E : Absorbing state (once entered, never leave).

Let X_n denote the state at time $n \geq 0$.

- (a) Write the transition matrix P .
- (b) Let $T = \min\{n \geq 0 : X_n = E\}$ be the hitting time of E . Define $k_X = \mathbb{E}[T \mid X_0 = X]$ for $X \in \{A, B, C, D\}$ (with $k_E = 0$). Write equations for k_A, k_B, k_C, k_D and solve for k_A .
- (c) Define $h_X = \mathbb{P}(\text{hit } E \text{ before returning to } A \mid X_0 = X)$ for $X \in \{B, C, D\}$. Write equations for h_B, h_C, h_D and solve them.

- (d) Define $u_X = \mathbb{P}(\text{hit } D \text{ before } E \mid X_0 = X)$ for $X \in \{A, B, C\}$. Write equations for u_X and solve for u_A .

Solution.

- (a) The transition matrix is given by

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 \\ 1/3 & 0 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 0 & 1/4 & 1/4 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Rows/columns in order A, B, C, D, E .

- (b) We have

$$\begin{cases} k_A = 1 + \frac{1}{2}k_B + \frac{1}{2}k_C \\ k_B = 1 + \frac{1}{3}k_A + \frac{1}{3}k_C + \frac{1}{3}k_D \\ k_C = 1 + \frac{1}{4}k_A + \frac{1}{4}k_B + \frac{1}{4}k_D \\ k_D = 1 + \frac{1}{2}k_B \end{cases}$$

Solving yields $k_A = \frac{113}{16}$.

- (c) Let $h_A = 0$ in equations because hitting A before E means failure. Then

$$\begin{cases} h_B = \frac{1}{3} \cdot 0 + \frac{1}{3}h_C + \frac{1}{3}h_D \\ h_C = \frac{1}{4} \cdot 0 + \frac{1}{4}h_B + \frac{1}{4}h_D + \frac{1}{4} \cdot 1 \\ h_D = \frac{1}{2}h_B + \frac{1}{2} \cdot 1 \end{cases}$$

Solving yields $h_B = \frac{7}{17}$, $h_C = \frac{9}{17}$ and $h_D = \frac{12}{17}$.

- (d) Let $u_D = 1, u_E = 0$. We have

$$\begin{cases} u_A = \frac{1}{2}u_B + \frac{1}{2}u_C \\ u_B = \frac{1}{3}u_A + \frac{1}{3}u_C + \frac{1}{3} \\ u_C = \frac{1}{4}u_A + \frac{1}{4}u_B + \frac{1}{4} \end{cases}$$

Solving yields $u_A = \frac{9}{13}$.

Problem 3. Specify the classes of the following Markov chains, and determine whether they are transient or recurrent:

$$\begin{aligned} \text{(a) } \mathbf{P}_1 &= \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} & \text{(b) } \mathbf{P}_2 &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ \text{(c) } \mathbf{P}_3 &= \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} & \text{(d) } \mathbf{P}_4 &= \begin{pmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

Solution.

- (a) $\{0, 1, 2\}$ recurrent.
- (b) $\{0, 1, 2, 3\}$ recurrent.
- (c) $\{0, 2\}$ recurrent, $\{1\}$ transient, $\{3, 4\}$ recurrent.
- (d) $\{0, 1\}$ recurrent, $\{2\}$ recurrent, $\{3\}$ transient, $\{4\}$ transient.

Problem 4. Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other one.

- (a) What proportion of flips use coin 1?
- (b) If we start the process with coin 1, what is the probability that coin 2 is used on the fifth flip?

Solution. If the state at time n is the k -th coin to be flipped, where $k \in \{1, 2\}$, then the process constitutes a two-state Markov chain with transition probabilities:

$$P_{1,1} = 0.6 = 1 - P_{1,2}, \quad P_{2,1} = 0.5 = 1 - P_{2,2}$$

where state 1 represents flipping coin 1 and state 2 represents flipping coin 2.

- (a) The stationary probabilities satisfy:

$$\begin{aligned}\pi_1 &= \pi_1 P_{1,1} + \pi_2 P_{2,1} \\ \pi_2 &= \pi_1 P_{1,2} + \pi_2 P_{2,2} \\ 1 &= \pi_1 + \pi_2\end{aligned}$$

Solving yields $\pi_1 = 5/9$, $\pi_2 = 4/9$. So the proportion of flips that use coin 1 is $5/9$.

- (b) We compute $P_{1,2}^{(4)} = 0.4444$.

Problem 5. A flea moves around the vertices of a triangle in the following manner: Whenever it is at vertex i it moves to its clockwise neighbor vertex with probability p_i and to the counterclockwise neighbor with probability $q_i = 1 - p_i$, $i = 1, 2, 3$.

- (a) Find the proportion of time that the flea is at each of the vertices.
- (b) How often does the flea make a counterclockwise move that is then followed by five consecutive clockwise moves?

Solution. States: vertices 1, 2, 3 arranged clockwise. From vertex i :

- Clockwise to $i + 1 \pmod{3}$: probability p_i
- Counterclockwise to $i - 1 \pmod{3}$: probability $q_i = 1 - p_i$

$$P = \begin{pmatrix} 0 & p_1 & 1 - p_1 \\ 1 - p_2 & 0 & p_2 \\ p_3 & 1 - p_3 & 0 \end{pmatrix}$$

(a) The stationary probabilities π_i , $i = 1, 2, 3$, satisfy:

$$\begin{aligned}\pi_1 &= q_2\pi_2 + p_3\pi_3 \\ \pi_2 &= p_1\pi_1 + q_3\pi_3 \\ 1 &= \pi_1 + \pi_2 + \pi_3\end{aligned}$$

(b) For convenience, define $p_0 := p_3$, $p_4 := p_1$, $p_5 := p_2$, $p_6 := p_3$. The proportion of time that there is a counterclockwise move from a state i that is followed by 5 clockwise moves is:

$$\pi_i q_i p_{i-1} p_i p_{i+1} p_{i+2} p_{i+3}$$

So the answer to (b) is:

$$\sum_{i=1}^3 \pi_i q_i p_{i-1} p_i p_{i+1} p_{i+2} p_{i+3}$$

Problem 6. A weather system follows these rules each day:

- If it rained on **all** of the past three days, it will rain today with probability 0.8.
- If it rained on **none** of the past three days, it will rain today with probability 0.2.
- In **all other cases**, the weather today is the same as yesterday's weather with probability 0.6, and opposite with probability 0.4.

Define a Markov chain where the state represents the weather pattern of the past three days. Determine the transition probability matrix $\mathbf{P}$ for this chain.

Solution. Define the state as the weather pattern over the past three days, using R for rain and D for dry (no rain). The 8 possible states are:

- RRR (rain all three past days)
- RRD (rain, rain, dry)
- RDR (rain, dry, rain)
- RDD (rain, dry, dry)
- DRR (dry, rain, rain)
- DRD (dry, rain, dry)
- DDR (dry, dry, rain)
- DDD (dry all three past days)

The transition rules:

- If state 1 (RRR): rain today with probability 0.8, dry with 0.2
- If state 8 (DDD): rain today with probability 0.2, dry with 0.8
- Otherwise: today's weather equals yesterday's with probability 0.6

When transitioning to the next state, we shift the history: drop the oldest day, append today's weather. The transition matrix $\mathbf{P}$ is:

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.4 & 0.6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.6 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.6 & 0.4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.6 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0.8 \end{bmatrix}.$$

Problem 7. Let X_n be the number of families that check into a hotel on day n , where $\{X_n\}_{n \geq 1}$ are independent $\text{Poisson}(\lambda)$ random variables. Each family that checks in stays for a random number of days that follows a geometric distribution with parameter p , ($P(\text{stay} = k) = p(1-p)^{k-1}$ for $k \geq 1$), with all stay durations independent of each other and of the arrivals process. Let Y_n be the number of families present in the hotel at the *beginning* of day n (before new arrivals on day n).

- (a) Show that $\{Y_n\}_{n \geq 1}$ is a Markov chain. Identify the state space and give an intuitive explanation.
- (b) Given $Y_n = i$, find the distribution of Y_{n+1} . That is, determine the transition probabilities $P_{ij} = P(Y_{n+1} = j \mid Y_n = i)$.

Solution.

- (a) Markov Property

The state $S = \{0, 1, 2, \dots\}$ is the possible number of families in the hotel. Given $Y_n = i$ (number of families at beginning of day n), let D be the number of these families that *depart* during day n . Since each family's stay duration is geometric(p) starting at 1, the probability that a given family departs on any particular day (given they are still present) is p , independently across families. Thus:

$$D \mid Y_n = i \sim \text{Binomial}(i, p)$$

Let $A_n = X_n$ be the new arrivals on day n , where $A_n \sim \text{Poisson}(\lambda)$, independent of everything. Then we have

$$Y_{n+1} = (Y_n - D) + A_n.$$

Observe that given Y_n , the distribution of Y_{n+1} depends only on:

- 1. $D \sim \text{Binomial}(Y_n, p)$ (departures from existing families)
- 2. $A_n \sim \text{Poisson}(\lambda)$ (new arrivals), independent of D

Thus, given Y_n , Y_{n+1} is conditionally independent of $Y_{n-1}, Y_{n-2}, \dots$. Hence $\{Y_n\}$ is a Markov chain.

- (b) For $i, j \geq 0$:

$$P_{ij} = P(Y_{n+1} = j \mid Y_n = i) = \sum_{d=0}^i P(D = d \mid Y_n = i) \cdot P(A_n = j - i + d)$$

But $P(A_n = k) = 0$ if $k < 0$. Since $a = j - i + d$ must be ≥ 0 , we require:

$$d \geq i - j$$

Also $0 \leq d \leq i$. Hence d runs from $\max(0, i - j)$ to i . Thus:

$$P_{ij} = \sum_{d=\max(0, i-j)}^i \binom{i}{d} p^d (1-p)^{i-d} \cdot e^{-\lambda} \frac{\lambda^{j-i+d}}{(j-i+d)!}$$